



USAMA

Palu, Central Sulawesi, Indonesia

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Bachelor of Informatics Engineering from Universitas Tadulako passionate about AI-powered systems, Embedded Systems, and Software Development. Experienced in developing intelligent automation and hardware-software integration projects using modern technologies.

Technical skills

- **Frontend Development** (HTML, CSS, Tailwind CSS, React JS, Next JS, React Native, TypeScript)
- **Backend Development** (Node JS, MySQL Database)
- **Programming Languages** (C/C++, Python, JavaScript, TypeScript)
- **Embedded Systems & Hardware** (Embedded Systems, IoT, Robotics, PCB Design, Electrical Schematic Design)
- **Artificial Intelligence** (AI, Machine Learning, Deep Learning Modeling)
- **Design & Multimedia** (Visual Design, Multimedia Editing)
- **Office Applications** (Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Google Docs, Google Sheets)
- **Version Control** (Git/GitHub)

Tools

- **Development & Engineering Tools** (Arduino IDE, Eagle, EasyEDA, Fritzing, Proteus, Wokwi, Tinkercad, Visual Studio Code, PyCharm, Postman)
- **IoT & Cloud Platforms** (ThingSpeak, Node-RED, Blynk, ThingsBoard, Google Colab)
- **Design & Multimedia Tools** (Adobe Photoshop, Adobe Illustrator, Adobe After Effects, Adobe Premiere Pro, CorelDRAW, Figma, Canva)

Soft Skills

- Analytical Thinking
- Teamwork
- Communication
- Problem Solving
- Interpersonal Skills
- Leadership
- Adaptability
- Independent Work

Projects

Monitoring and controlling irrigation gates (ESP32, PHP, MySQL, JavaScript)

- Developed IoT system for real-time monitoring and remote control of irrigation gates
- Implemented ESP32-based data acquisition for water level and gate status
- Designed web dashboard for visualization and control interface
- Enabled efficient irrigation management by reducing manual monitoring

Euforia - PCB Defect Detection System (Python, Computer Vision, Deep Learning)

- Developed AI system for detecting PCB defects using image processing techniques
- Built a detection model to identify defective PCB boards

MP4 HD – Patient Monitor (Hardware Engineering, IoT)

- Developed a patient monitoring system for real-time health data observation
- Implemented hardware-based solution integrating sensors and microcontroller
- Completed as a final project for the MSIB6 Stechoq Hardware Engineering program

Landing Page – Pramuka SMANSA Poso (React.js, Tailwind)

- Developed responsive landing page using React.js
- Applied modern UI/UX principles for better user experience
- Contributed as a voluntary project for the organization

Plant Monitoring System (IoT, ESP32, Sensors, Blynk)

- Developed an IoT-based system for real-time plant monitoring
- Integrated sensors with ESP32 for environmental data collection
- Utilized Blynk platform to build a real-time monitoring dashboard
- Delivered a reliable solution to support plant research and client requirements

Mobile Monitoring Application (React Native)

- Developed a mobile application for monitoring and controlling environmental parameters
- Implemented features to track temperature, humidity, soil pH, water quality, and lighting systems
- Designed user-friendly interface for real-time monitoring and control
- Delivered solution tailored to meet client requirements

Asset Inventory and Management Portal (React.js, TypeScript, Express.js, MySQL)

- Developed a fullstack web-based asset management system
- Built frontend using React.js and TypeScript for responsive and scalable UI
- Developed RESTful API using Express.js for data processing and integration
- Implemented asset inventory, tracking, and management features
- Improved transparency and efficiency in municipal asset administration
- Delivered solution for Palu City Environmental Department

Automated Paint Can Defect Detection & Sorting System (YOLOv12, Python, OpenCV, Arduino, Embedded Systems)

- Developed an end-to-end automated inspection and sorting system for paint can defect detection using YOLOv12 and conveyor belt integration
- Built a real-time object detection system using Python, OpenCV, and YOLOv12 with multi-camera configuration to reduce blind spots
- Designed and implemented conveyor belt mechanical systems, electronic circuits, and PCB layouts for automated industrial inspection
- Integrated sensors, microcontrollers, DC motors, and servo actuators for synchronized detection and automatic sorting processes
- Trained and optimized deep learning models using 1,200+ annotated image datasets with augmentation techniques for robust detection performance
- Achieved high detection performance with 0.98 Precision, 0.96 Recall, 0.97 F1-Score, and 0.96 mAP50-95
- Developed real-time monitoring GUI for live detection and operational control using OpenCV
- Conducted system performance evaluation based on sorting accuracy, latency, throughput, conveyor speed, and lighting conditions
- Implemented embedded control systems using Arduino IDE for conveyor and actuator automation
- Designed as a smart industrial automation solution for quality control and automated sorting systems

Experiences

05 November 2022

Jury in the Robotics Event PERMIKOMNAS REGION X GORONTALO

Februari 2021 – June 2023

Teaching Assistant / Bachelor of Informatics Engineering Program, Tadulako University

- Assisted in the preparation and management of course materials, assignments, and examinations
- Provided academic guidance to students regarding practicum materials and assignments
- Collaborated with lecturers to support the smooth implementation of practical sessions
- Conducted tutoring and additional academic assistance sessions for students

November 2024 – December 2024

Teaching Assistant / Bachelor of Nursing Program, Widya Nusantara University

- Prepared and managed practicum materials and assignments
- Provided academic guidance to students related to Information Technology subjects

September 2024 – April 2025

Chairperson / Animedia Tadulako

- Led, managed, and developed the organization
- Designed and organized work programs and organizational activities
- Managed organizational operations efficiently
- Maintained member productivity and organizational quality
- Established and maintained good relationships with external parties
- Represented and advocated for organizational interests while building a positive reputation

April 2025 – May 2025

IT Staff / Environmental Agency of Palu City

- Developed a web-based Asset Inventory and Management Portal for the Environmental Agency of Palu City

Education Level

2020 - 2026

Universitas Tadulako - Palu, Central Sulawesi

Bachelor Degree in Informatics Engineering, 3.71/4.00

August - December 2023

Study Independent Artificial Intelligence at Startup Campus

- AI Foundation
- Python Foundation
- Data Science
- Machine Learning and Deep Learning
- Computer Vision
- Git Collaboration

February - June 2024

Study Independent Hardware Engineering at PT. Stechoq

- Internet of Things
 - Sensors dan Actuators
 - Programmable Logic Controller (PLC) Supervisory Control and Data Acquisition (SCADA)
 - RFID Technology, Barcode and QR Code
 - Industrial Control System
 - Electronic Design
 - Microcontroller
 - Wireless Communication Technology
 - Analog and digital Electronic
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